

DAY 5 SQL- CONSTRAINTS AND JOINS

USE slatask;

```
CREATE TABLE Customers (  
    CustomerID INT PRIMARY KEY,  
    CustomerName VARCHAR(50),  
    City VARCHAR(50)  
);
```

```
CREATE TABLE Products (  
    ProductID INT PRIMARY KEY,  
    ProductName VARCHAR(50),  
    Price DECIMAL(10,2)  
);
```

```
CREATE TABLE Orders (  
    OrderID INT PRIMARY KEY,  
    CustomerID INT,  
    ProductID INT,  
    OrderDate DATE,  
    Quantity INT,  
    FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID),  
    FOREIGN KEY (ProductID) REFERENCES Products(ProductID)  
);
```

-- INSERT DATA

```
INSERT INTO Customers VALUES  
(1,'Alice','New York'),  
(2,'Bob','London'),  
(3,'Charlie','Paris'),  
(4,'David','Berlin');
```

```
INSERT INTO Products VALUES
```

```
(101,'Laptop',1200.00),
```

```
(102,'Mouse',25.00),
```

```
(103,'Keyboard',50.00);
```

```
INSERT INTO Orders VALUES
```

```
(1,1,101,'2023-01-01',1),
```

```
(2,2,102,'2023-01-02',2),
```

```
(3,1,103,'2023-01-03',1),
```

```
(4,NULL,101,'2023-01-04',1);
```

```
-- CONSTRAINT (existing tables)
```

```
-- ADD UNIQUE constraint
```

```
ALTER TABLE Customers ADD CONSTRAINT uq_city UNIQUE(City);
```

```
-- DROP UNIQUE constraint
```

```
ALTER TABLE Customers DROP INDEX uq_city;
```

```
-- ADD composite PRIMARY KEY example
```

```
ALTER TABLE Orders DROP PRIMARY KEY;
```

```
ALTER TABLE Orders ADD PRIMARY KEY (OrderID, ProductID);
```

```
-- DROP composite PRIMARY KEY and restore
```

```
ALTER TABLE Orders DROP PRIMARY KEY;
```

```
ALTER TABLE Orders ADD PRIMARY KEY (OrderID);
```

```
-- ADD FOREIGN KEY manually
```

```
ALTER TABLE Orders
```

```
ADD CONSTRAINT fk_customer
```

```
FOREIGN KEY(CustomerID) REFERENCES Customers(CustomerID);
```

-- DROP FOREIGN KEY

ALTER TABLE Orders DROP FOREIGN KEY fk_customer;

OUTPUT

125	17:09:11	CREATE TABLE Products (ProductID INT PRIMARY KEY, ProductName VARCHAR(50), Price DEC...	0 row(s) affected
126	17:09:17	CREATE TABLE Orders (OrderID INT PRIMARY KEY, CustomerID INT, ProductID INT, OrderDate ...	0 row(s) affected
127	17:09:36	INSERT INTO Customers VALUES (1,'Alice','New York'), (2,'Bob','London'), (3,'Charlie','Paris'), (4,'David','Berlin')	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0
128	17:09:40	INSERT INTO Products VALUES (101,'Laptop',1200.00), (102,'Mouse',25.00), (103,'Keyboard',50.00)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
129	17:09:44	INSERT INTO Orders VALUES (1,1,101,'2023-01-01',1), (2,2,102,'2023-01-02',2), (3,1,103,'2023-01-03',1), (4,...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0
130	17:10:13	ALTER TABLE Customers ADD CONSTRAINT uq_city UNIQUE(City)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
131	17:10:31	ALTER TABLE Customers DROP INDEX uq_city	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
132	17:10:46	ALTER TABLE Orders DROP PRIMARY KEY	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0
133	17:10:53	ALTER TABLE Orders ADD PRIMARY KEY (OrderID, ProductID)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
134	17:11:01	ALTER TABLE Orders DROP PRIMARY KEY	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0
135	17:11:11	ALTER TABLE Orders ADD PRIMARY KEY (OrderID)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
136	17:11:25	ALTER TABLE Orders ADD CONSTRAINT fk_customer FOREIGN KEY(CustomerID) REFERENCES Custome...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0
137	17:11:57	ALTER TABLE Orders DROP FOREIGN KEY fk_customer	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

-- JOIN PRACTICE

-- 1 CustomerName + OrderID for every order

SELECT Customers.CustomerName, Orders.OrderID

FROM Orders

INNER JOIN Customers

ON Orders.CustomerID = Customers.CustomerID;

OUTPUT:

	CustomerName	OrderID
▶	Alice	1
	Alice	3
	Bob	2

-- 2 All customers + OrderID (include no orders)

SELECT Customers.CustomerName, Orders.OrderID

FROM Customers

LEFT JOIN Orders

ON Customers.CustomerID = Orders.CustomerID;

OUTPUT

	CustomerName	OrderID
▶	Alice	1
	Alice	3
	Bob	2
	Charlie	NULL
	David	NULL

-- 3 All products + OrderID

```
SELECT Products.ProductName, Orders.OrderID
FROM Products
LEFT JOIN Orders
ON Products.ProductID = Orders.ProductID;
```

OUTPUT

	ProductName	OrderID
▶	Laptop	1
	Laptop	4
	Mouse	2
	Keyboard	3

-- 4 CustomerName, ProductName, OrderDate

```
SELECT Customers.CustomerName, Products.ProductName, Orders.OrderDate
FROM Orders
INNER JOIN Customers
ON Orders.CustomerID = Customers.CustomerID
INNER JOIN Products
ON Orders.ProductID = Products.ProductID;
```

Output

	CustomerName	ProductName	OrderDate
▶	Alice	Laptop	2023-01-01
	Bob	Mouse	2023-01-02
	Alice	Keyboard	2023-01-03

-- 5 Customers who bought Laptop

```
SELECT DISTINCT Customers.CustomerName
FROM Orders
JOIN Customers ON Orders.CustomerID = Customers.CustomerID
JOIN Products ON Orders.ProductID = Products.ProductID
WHERE Products.ProductName='Laptop';
```

Output

	CustomerName
▶	Alice

-- 6 Total amount spent by each customer

```
SELECT Customers.CustomerName,
SUM(Products.Price * Orders.Quantity) AS total_spent
FROM Orders
JOIN Customers ON Orders.CustomerID = Customers.CustomerID
JOIN Products ON Orders.ProductID = Products.ProductID
GROUP BY Customers.CustomerName;
```

Output

	CustomerName	total_spent
▶	Alice	1250.00
	Bob	50.00

-- 7 Orders without customer

```
SELECT * FROM Orders WHERE CustomerID IS NULL;
```

Output

	OrderID	CustomerID	ProductID	OrderDate	Quantity
▶	4	NULL	101	2023-01-04	1
*	NULL	NULL	NULL	NULL	NULL

-- 8 Cartesian product

```
SELECT Customers.CustomerName, Products.ProductName
FROM Customers
CROSS JOIN Products;
```

Output

	CustomerName	ProductName
▶	Alice	Keyboard
	Alice	Mouse
	Alice	Laptop
	Bob	Keyboard
	Bob	Mouse
	Bob	Laptop
	Charlie	Keyboard
	Charlie	Mouse
	Charlie	Laptop
	David	Keyboard
	David	Mouse
	David	Laptop

-- 9 Self join (Manager example)

```
ALTER TABLE Customers ADD ManagerID INT;
```

```
SELECT C.CustomerName AS Customer,
```

```
M.CustomerName AS Manager
```

```
FROM Customers C
```

```
LEFT JOIN Customers M
```

```
ON C.ManagerID = M.CustomerID;
```

Output

	Customer	Manager
▶	Alice	NULL
	Bob	NULL
	Charlie	NULL
	David	NULL

-- 10 New York customers ordered >1 quantity

```
SELECT Customers.CustomerName, Products.ProductName
```

```
FROM Orders
```

```
JOIN Customers ON Orders.CustomerID = Customers.CustomerID
```

```
JOIN Products ON Orders.ProductID = Products.ProductID
```

```
WHERE Customers.City='New York'
```

```
AND Orders.Quantity>1;
```

output

	CustomerName	ProductName